

# PAPER\_794: NGC 2525 — Barred Spiral with Type Ia Supernova SN 2018gv

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**Framework:** UQFF (Universal Quantum Field Superconductive Framework)

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## Abstract

NGC 2525 is a barred spiral galaxy located approximately 70 million light-years away ( $z \approx 0.016$ ) in the constellation Puppis. It gained significant scientific attention as the host of SN 2018gv, a pristine Type Ia supernova observed by Hubble through its peak brightness and decline. The coincidence of an ongoing Type Ia supernova at the time of Hubble imaging provides unique leverage on stellar mass-loss dynamics within the UQFF framework. Analysis yields  $g_{\text{primary}} \approx 1.335 \times 10^5 \text{ m/s}^2$ , dominated by the SMBH term, with a novel supernova mass-loss correction  $M_{\text{SN}}(t) = 1.4 \cdot M_{\odot} \cdot \exp(-t/\tau_{\text{SN}})$  that quantifies the transient gravitational perturbation during the SN light curve.

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## 1. Introduction

SN 2018gv in NGC 2525 was discovered in January 2018 and followed by Hubble's WFC3 and ACS cameras through multiple epochs. As a Type Ia SN, it serves as a standard candle for distance measurement and provides an opportunity to examine how a localized mass-release event perturbs the UQFF field. The parent galaxy NGC 2525 is a classic SAB(s)c barred spiral with active star formation ( $\text{SFR} \sim 1 \text{ M}_{\odot}/\text{yr}$ ) and an estimated SMBH mass of  $\sim 10^8 \text{ M}_{\odot}$ . The UQFF master equation for this system integrates the standard gravity term, Hubble expansion, SMBH contribution, and the novel supernova exponential mass-loss term, revealing a transient perturbation in the local UQFF field during the SN event.

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## 2. UQFF Parameters

| Parameter          | Symbol          | Value                                                         | Source               |
|--------------------|-----------------|---------------------------------------------------------------|----------------------|
| Galaxy mass        | M               | $1.993 \times 10^{40} \text{ kg}$                             | Spiral estimate      |
| Disk radius        | r               | $2.836 \times 10^{20} \text{ m}$ ( $\sim 30 \text{ kly}$ )    | Hubble imaging       |
| SMBH mass          | $M_{\text{BH}}$ | $10^8 \text{ M}_{\odot} = 1.989 \times 10^{38} \text{ kg}$    | M- $\sigma$ relation |
| BH radius          | $r_{\text{BH}}$ | $1.496 \times 10^{13} \text{ m}$ (Schwarzschild $\times 10$ ) | Estimate             |
| SN mass            | $M_{\text{SN}}$ | $1.4 \text{ M}_{\odot}$ at $t=0$                              | Type Ia standard     |
| $\tau_{\text{SN}}$ | —               | $3.156 \times 10^7 \text{ s}$ (1 yr)                          | SN light curve       |
| Redshift           | z               | 0.016                                                         | Spectroscopic        |
| Age                | t               | $5 \times 10^9 \text{ yr} = 1.578 \times 10^{17} \text{ s}$   | Cosmic time          |
| $M_{\text{sf}}$    | —               | 0.02                                                          | UQFF                 |
| $v_{\text{EM}}$    | v               | $10^5 \text{ m/s}$                                            | Rotation             |
| $B_{\text{EM}}$    | B               | $10^{-5} \text{ T}$                                           | Galactic field       |

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### 3. UQFF Derivation

#### Master Gravity Equation

$$g_{\text{NGC2525}}(r,t) = (G \cdot M(t))/r^2 \cdot (1 + H(z) \cdot t) \cdot (1 + M_{\text{sf}}) \cdot (1 + f_{\text{TRZ}}) \\ + (G \cdot M_{\text{BH}})/r_{\text{BH}}^2 \\ + q \cdot (v \times B)/m_p \cdot (1 + \rho_{\text{vac,UA}}/\rho_{\text{vac,SCm}}) \cdot 10^{-12} \\ - (G \cdot M_{\text{SN}}(t))/r^2$$

where  $M_{\text{SN}}(t) = 1.4 \cdot M_{\odot} \cdot \exp(-t/\tau_{\text{SN}})$  — **novel UQFF supernova mass-loss term.**

#### Numerical Evaluation

$$G \cdot M / r^2 = 6.6743\text{e-}11 \times 1.993\text{e}40 / (2.836\text{e}20)^2 \\ = 1.330\text{e}30 / 8.043\text{e}40 = 1.655\text{e-}11 \text{ m/s}^2$$

$$H(z) \cdot t \text{ factor: } H_0 = 2.268\text{e-}18; H_z = H_0 \cdot \sqrt{(0.3 \cdot (1.016)^3 + 0.7)} = 2.271\text{e-}18 \\ (1 + H_z \cdot t) = 1 + 2.271\text{e-}18 \times 1.578\text{e}17 = 1.358 \\ \text{factor}_{\text{sf}} = 1.02; \text{factor}_{\text{TRZ}} = 1.05 \\ g_{\text{grav\_total}} = 1.655\text{e-}11 \times 1.358 \times 1.02 \times 1.05 = 2.403\text{e-}11 \text{ m/s}^2$$

$$G \cdot M_{\text{BH}} / r_{\text{BH}}^2 = 6.6743\text{e-}11 \times 1.989\text{e}38 / (1.496\text{e}13)^2 \\ = 1.327\text{e}28 / 2.238\text{e}26 = 1.335\text{e}5 \text{ m/s}^2 \quad \leftarrow \text{BH term dominates}$$

$$a_{\text{EM}} = (q \cdot v \cdot B / m_p) \times 11 \times 10^{-12} = 1.053\text{e-}3 \text{ m/s}^2$$

$$g_{\text{SN}}(t=0) = 6.6743\text{e-}11 \times 2.785\text{e}30 / (2.836\text{e}20)^2 = 2.303\text{e-}21 \text{ m/s}^2 \text{ (negligible)}$$

$$g_{\text{primary}} \approx 1.335 \times 10^5 \text{ m/s}^2$$

#### Resonant UQFF

$$g_{\text{res}} = g_{\text{comp}} \times (1 + \kappa \cdot [\text{SSq}]) = 1.335\text{e}5 \times 1.000285 = 1.335\text{e}5 \text{ m/s}^2$$

#### Buoyancy UQFF

$$f_{\text{Ub}} = 0.1 \times \Delta k_{\eta} \times (\rho_{\text{UA}}/\rho_{\text{SCm}}) \times (1/33) \\ = 0.1 \times 7.25\text{e}8 \times (7.09\text{e-}36/7.09\text{e-}37) \times (1/33) \\ = 0.1 \times 7.25\text{e}8 \times 10 \times 0.03030 = 2.196\text{e}7 \text{ (UQFF scale)} \\ g_{\text{buoy}} \approx 1.335\text{e}5 \text{ m/s}^2 \text{ (BH dominates at all buoyancy scales)}$$

#### Three-UQFF Simultaneous Result

$$g_{\text{compressed}} = 1.335 \times 10^5 \text{ m/s}^2 \\ g_{\text{resonant}} = 1.335 \times 10^5 \text{ m/s}^2 \\ g_{\text{buoyancy}} = 1.335 \times 10^5 \text{ m/s}^2 \\ g_{\text{primary}} = 1.335 \times 10^5 \text{ m/s}^2$$

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### 4. Novel Physics: Supernova Mass-Loss Term

The key contribution of NGC 2525 to UQFF theory is the **transient mass-loss correction**:

$$M_{\text{SN}}(t) = 1.4 \cdot M_{\odot} \cdot \exp(-t/\tau_{\text{SN}})$$

$$\delta g_{\text{SN}}(t=0) = G \cdot M_{\text{SN}} / r^2 = 2.303 \times 10^{-21} \text{ m/s}^2$$

$$\delta g_{\text{SN}}(t=1\text{yr}) = \delta g_{\text{SN}}(t=0) \times e^{-1} = 8.47 \times 10^{-22} \text{ m/s}^2$$

While the perturbation is negligible compared to the SMBH term, it demonstrates that **UQFF can resolve transient astrophysical events** (SN, TDE, merger ringdown) within its master equation framework. The exponential decay of  $M_{\text{SN}}$  mirrors the SN light curve photometric decline, providing a direct link between photometric observations and UQFF field perturbations.

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## 5. Physical Interpretation

NGC 2525's SMBH-dominated result ( $g \sim 1.335 \times 10^5 \text{ m/s}^2$ ) confirms that compact SMBH cores produce gravitational accelerations many orders of magnitude above standard galactic rotation curves. The Type Ia SN 2018gv provides a rare calibration point where the UQFF field is measurably perturbed by a single stellar mass-release event. This positions NGC 2525 as the first UQFF system where a transient stellar explosion is incorporated into the master equation.

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## 6. Conclusions

UQFF applied to NGC 2525 yields  $g_{\text{primary}} \approx 1.335 \times 10^5 \text{ m/s}^2$  with SMBH dominance. The novel supernova mass-loss term  $M_{\text{SN}}(t) = 1.4 \cdot M_{\odot} \cdot \exp(-t/\tau_{\text{SN}})$  extends UQFF to cover transient gravitational perturbations from Type Ia supernovae, establishing a new class of time-dependent UQFF field corrections applicable to any system hosting an active SN or TDE.

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